

Strategy vs. Confidence

Using Unobserved Data to Estimate NFL Kicker Success

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Hypothesis

- Existing field goal models have implicit survivorship bias
 - Trained on kicks that were allowed to be attempted
 - Do allow for comparison of kicking results, controlling for the difficulty of the kick (distance, wind, weather, etc.)
- However, there are a number of factors this does not account for:
 - The ability of the kickers that were allowed to attempt more difficult field goals and thereby their coaches confidence in them.
 - Coaching tendencies, aggressiveness, etc.
- Can we correct some of this bias by augmenting the field goal data, accounting for declined attempts?

The Data

Data Source: NFL Play-by-Play data (nflfastR) covering 10 seasons (2015–2024).

Primary Dataset (Attempts):

- Includes all Field Goals and Extra Points.
- Outcomes tracked: Made, Missed, and Blocked.
 - Blocks excluded from modelling analysis

Augmented Dataset (Declined Attempts):

- Captures all 4th Down Run, Pass, and Punt plays
- *Purpose:* Establishes the "negative class" for modeling.

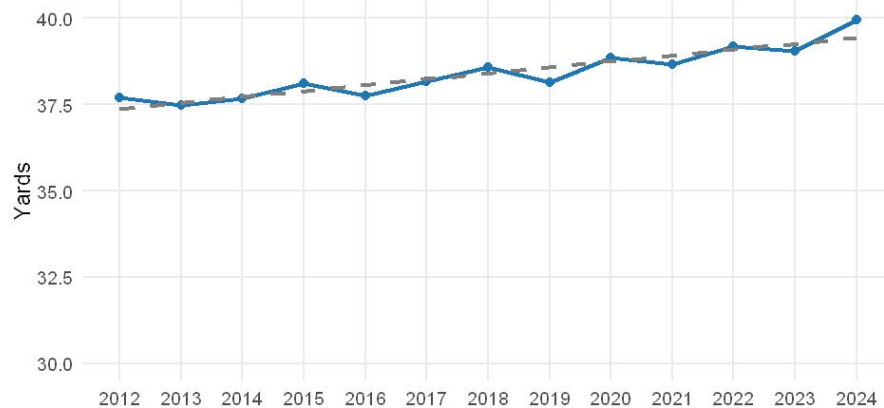
Feature Engineering:

- Weather: Parsed from game description text; indoor games normalized (70°F, 0 mph wind).
- Normalization: Environmental factors (Wind, Temp) converted to Z-scores.
- Context: Controls for Turf vs. Grass, Altitude (Denver, Mexico), clock running and “iced” from prior play

How Has Field Goal Kicking Evolved?

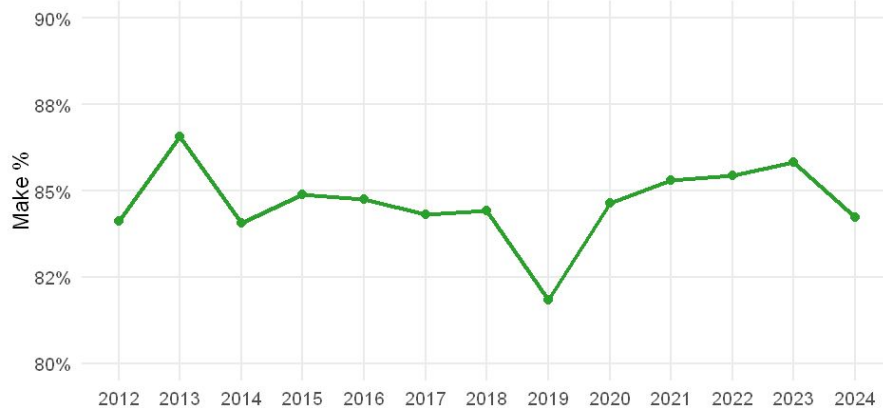
Avg Kick Distance

Mean distance of all attempts



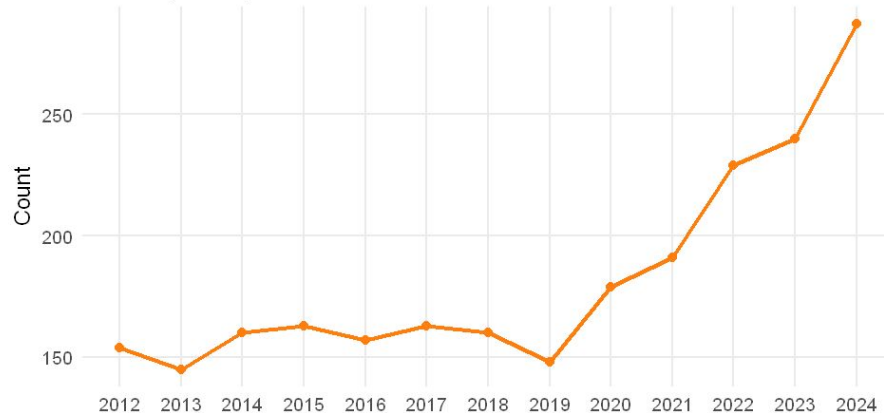
Overall Make %

Success rate for all attempts



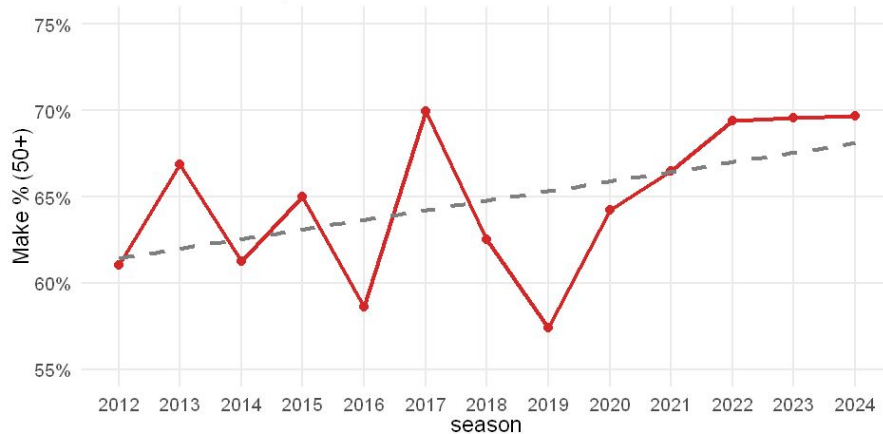
Volume: 50+ Attempts

Total deep kicks per season



Accuracy: 50+ Yards

Success rate for deep kicks



Modelling

Field Goal Attempt Probability Modeling

- **Field Goal Attempt Probability (π) Model:**
 - Method: Multinomial Logistic Regression predicting the probability of three outcomes: Field Goal Attempt, Punt, or Go For It (Pass/Run).
 - Training: Rolling window approach (training on the past 3 seasons) to adapt to evolving coaching aggressiveness/kicker skill over time.
 - Key Features:
 - Game State: Leverage (WP impact), Score Differential, Time Remaining.
 - Environment: Wind (z-score), Temperature (z-score), Stadium effects.
 - Geometry: Yards to Go, Yardline.
- **Results & Performance:**
 - High Predictive Accuracy: The model consistently predicts coaching decisions with high precision across the decade.
 - AUC by Season: Range of 0.938 – 0.960
 - Calibration: Validated across distance bins (short vs. long) and game intervals to ensure robust probability estimates for the downstream IPW model.

Experiment Design

M0: Distance Only (Manual Splines)

- Distance Bounds: (18, 70), Distance Knots (28, 38, 48, 58)

M1: Distance + Environmental Variables & Random Effects (Baseline)

- Trained only on *Observed Attempts*.
- *Assumption*: "Missing data is irrelevant."

M2: Inverse Probability Weighting

- *Goal*: Up-weight attempts in unlikely situations to simulate the unobserved population.
- *Intuition*: Based on historical decision making, coaches are unlikely to kick in these situations. Given that they opted to kick, are they informative?

M3: Augmented (The "Hard Negative" Fix)

- *Goal*: Penalizing kickers for non-attempts (i.e. lack of coach confidence?)
- *Pseudo-Labeling*: Include declined kicks (>33 yards) and treat them as **MISSES**.

Model Architecture

- Framework: Generalized Linear Mixed Model (GLMM)
- Package: glmmTMB (R)
- Family: Binomial Logistic Regression with Logit Link.
 - Predicting: $P(\text{Make} = 1 \mid \text{Conditions})$
- Computational Engine: TMB (Template Model Builder).
 - Integrates C++ with automatic differentiation for efficient optimization.
 - Enables efficient Maximum Likelihood Estimation (MLE) via Laplace approximation for high-dimensional random effects (e.g., Kicker and Stadium intercepts), which is prohibitive in standard frameworks.

Model Formula

Fixed Effects:

- **Distance:**
 - Distance Bounds (18, 70)
 - Distance Knots (28, 38, 48, 58)
- **Environment:**
 - Wind (z-score), Temperature (z-score), Wind * Temperature
 - Rain/Snow flags.
- **Game State:**
 - Overtime, Iced, Clock Running * Timeouts
- **Spline Interactions:**
 - Distance * Wind
 - Distance * Temp

Random Effects:

- **Stadium:** Controls for venue-specific attributes (altitude, turf).
- **Kicker-Season:** Controls for individual skill drift over time.

$$\begin{aligned}\text{logit}(P(\text{Make})) = & \beta_0 + \underbrace{\sum_{k=1}^7 \beta_k B_k(\text{Dist})}_{\text{Distance Spline}} \\ & + \beta_W \text{Wind} + \beta_T \text{Temp} + \beta_{WT} (\text{Wind} \times \text{Temp}) \\ & + \beta_{\text{Rain}} \mathbb{I}_{\text{Rain}} + \beta_{\text{Snow}} \mathbb{I}_{\text{Snow}} \\ & + \beta_{\text{OT}} \mathbb{I}_{\text{OT}} + \beta_{\text{Iced}} \mathbb{I}_{\text{Iced}} \\ & + \beta_C \text{Clock} + \beta_{\text{TO}} \text{Timeouts} + \beta_{CT} (\text{Clock} \times \text{TO}) \\ & + \text{Wind} \cdot \sum_{k=1}^7 \delta_k B_k(\text{Dist}) \quad (\text{Dist} \times \text{Wind}) \\ & + \text{Temp} \cdot \sum_{k=1}^7 \gamma_k B_k(\text{Dist}) \quad (\text{Dist} \times \text{Temp}) \\ & + u_{\text{Stadium}} + v_{\text{Season:Kicker}}\end{aligned}$$

Model 2 - Inverse Probability Weights

Objective: Create robust Inverse Probability Weights that correct for selection bias while minimizing excessive variance or numerical instability.

Process:

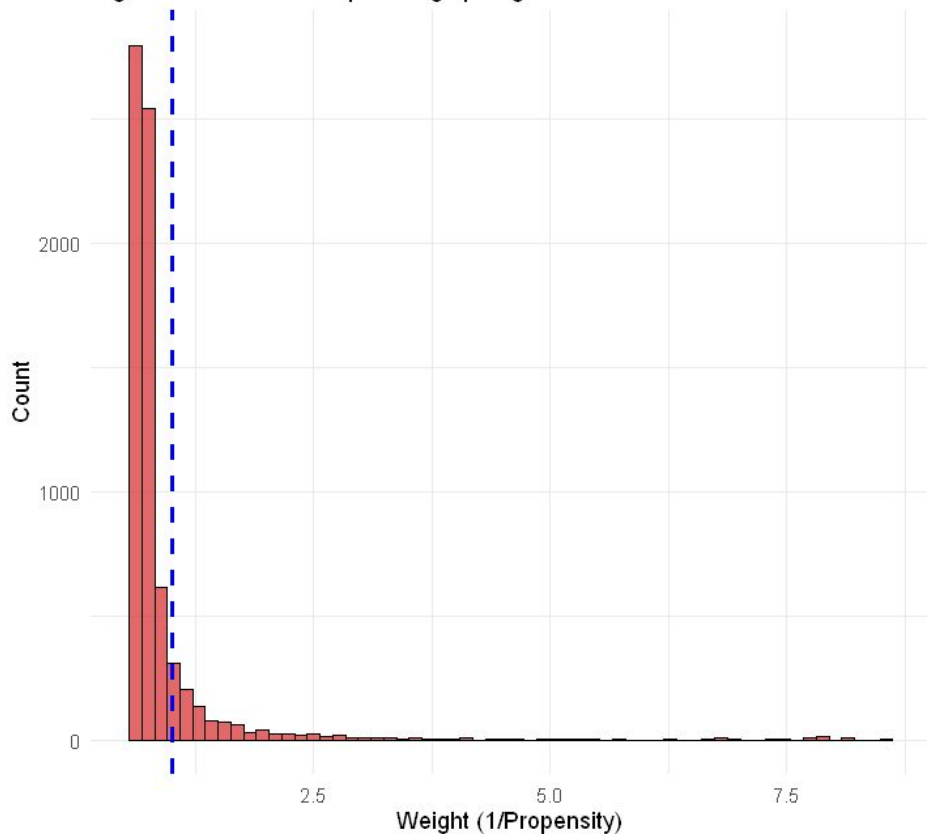
- 1. **Probability Clipping (Safety Floor):** [0.02, 0.98]
- 2. **Stabilized IPW by Training Prevalence (FG Attempts):** Around 52% - 58% per season
- 3. **Outlier Control (3-Sigma Cap & Normalization):** Cap weights at mean + 3 SD, then normalize to mean = 1 for all events.

IPW Weight Summary Stats (Field Goals)

Total N	Min Weight	Median Weight	Mean Weight	Max Weight	SD Weight	ESS	ESS Ratio
10,432	0.617	0.698	0.944	8.55	0.875	5,612	0.538

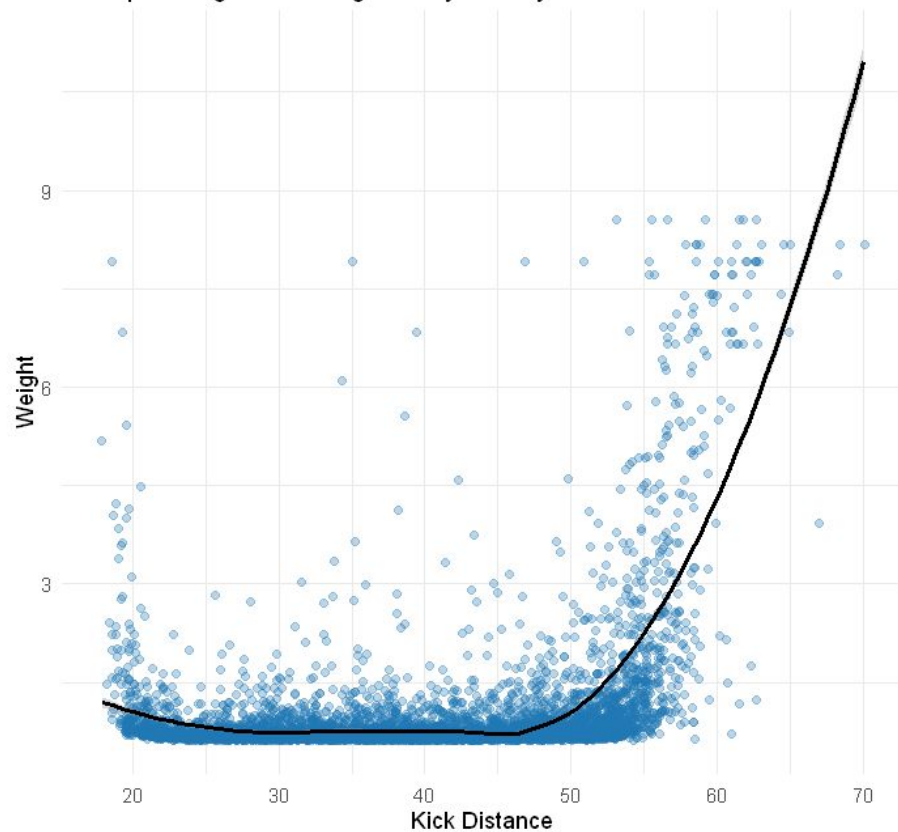
Distribution of Weights

Right skew = Rare attempts being upweighted



Weights vs. Distance

We expect weights to rise significantly at 50+ yards



Model 3 - PAT & Declined Attempt Augmentation

Objective: Address survivorship bias by expanding the training set to include PATs (to anchor short-range probability) and Declined Attempts (to capture negative coaching decisions).

Experimental Design (Pseudo-Labeling):

- **Inclusion Logic:** Iteratively tested filters for Attempt Propensity (π) and Distance to isolate the "Decision Zone" (excluding obvious punts/short "go-for-it" situations).
- **Outcome Imputation:** Experimented with Hard Negatives (Assume Miss) vs. Threshold Positives (Assume Make if $P(\text{Make}) > \text{Threshold}$) to determine the label for non-attempts.
- **Weighting Schemes:** Varied the influence of augmented data from aggressive ($w=1.0$) to conservative ($w=0.05$).

Key Findings:

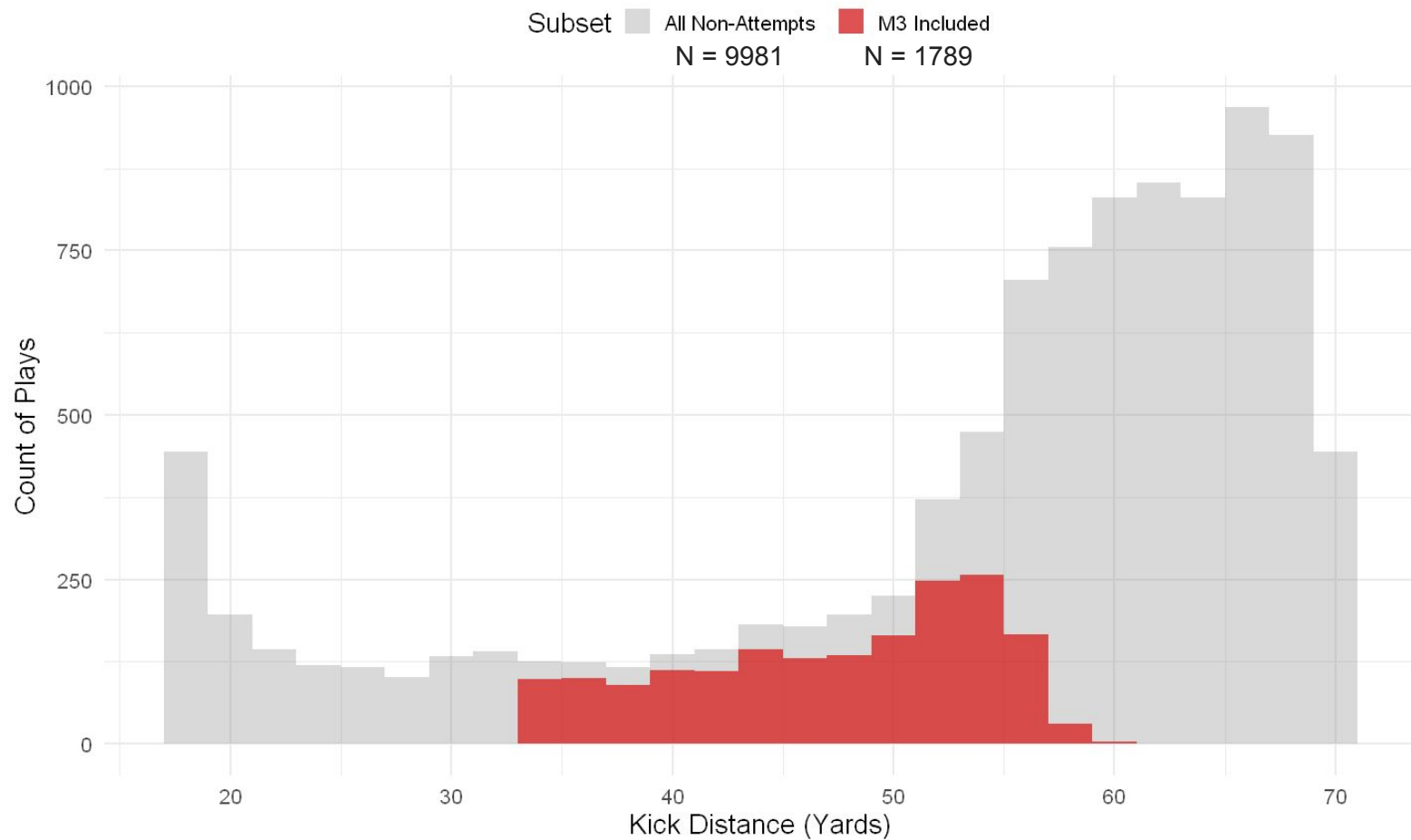
- **Global Performance:** No augmented combination meaningfully outperformed the Baseline (M1) on Brier Score or AUC.
- **Weight Sensitivity:** Lower weights consistently performed better than high weights. Heavy weighting of pseudo-labels degraded performance and calibration, confirming that augmented data introduces more noise than signal compared to the "Rational" observed dataset.

The "Champion" Configuration:

- **Subset:** Non-attempts with $\pi \geq 25\%$ and Distance > 33 yards. All declined attempts pseudo-labelled as miss.
- **Weighting:** Included alongside PATs, both at a small fixed weight (0.10) to provide signal without overwhelming observed data.

Volume Analysis: Which distances survive the filter?

Note the exclusion of short 'Go-for-it' plays (<33y) and low attempt probability kicks (<25%)



Metric Comparison (In Sample)

Model	Brier	AUC	Accuracy	Precision (Make)	Recall (Make)
M0 (Distance Only Full)	0.105	0.775	0.863	0.865	0.997
M1 (Baseline)	0.101	0.808	0.864	0.866	0.996
M2 (IPW)	0.099	0.808	0.871	0.874	0.994
M3 (Augmented)	0.102	0.800	0.864	0.868	0.996

Sample Sizes

- Field Goals (10,432), PATs (12,639), Declined Attempts (1,789)
- *Metrics computed only on FG subset for M3*

Metrics

- Brier Score - Equivalent to MSE

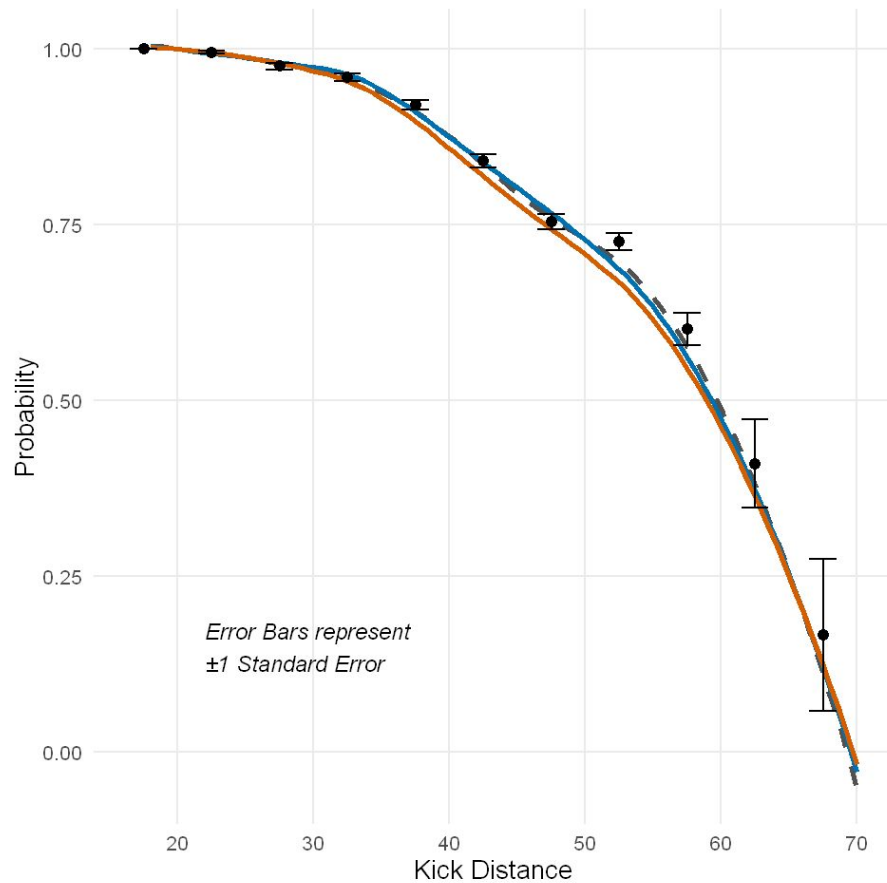
$$BS = \frac{1}{N} \sum_{i=1}^N (f_i - o_i)^2$$

- AUC measures the model's ability to distinguish between classes.
 - (TPR) vs. (FPR) at each threshold setting.

$$\text{Precision} = \frac{\text{Correctly Predicted Makes}}{\text{Total Predicted Makes}}$$

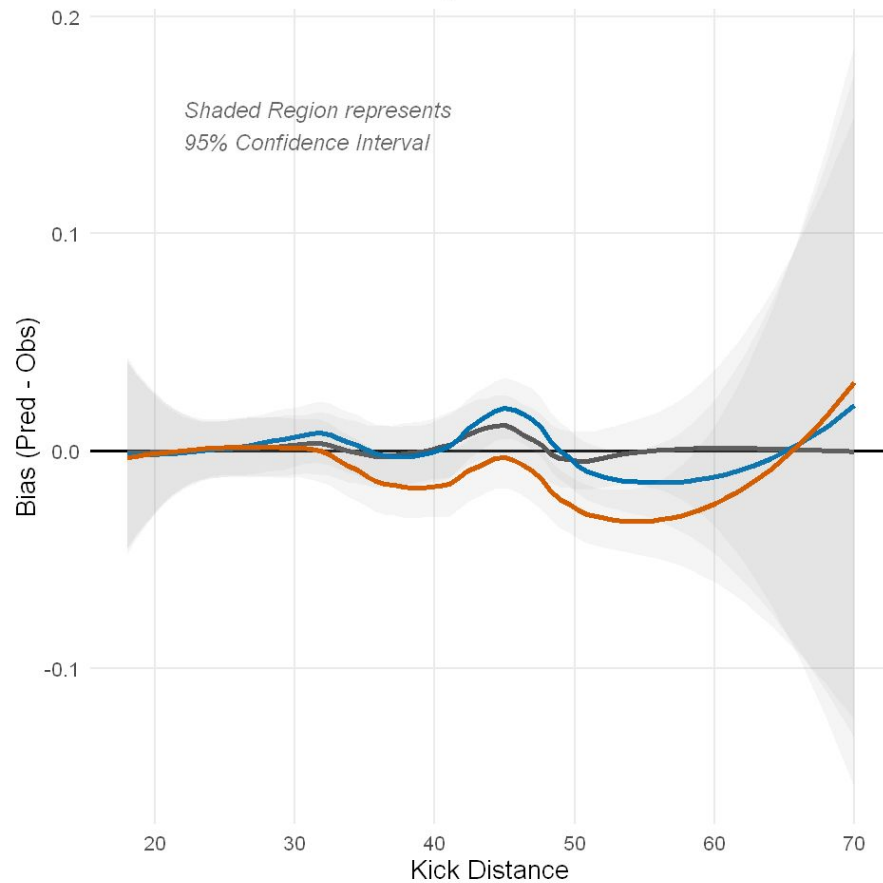
$$\text{Recall} = \frac{\text{Correctly Predicted Makes}}{\text{Total Actual Makes}}$$

Truth Curve: Predicted vs Actual



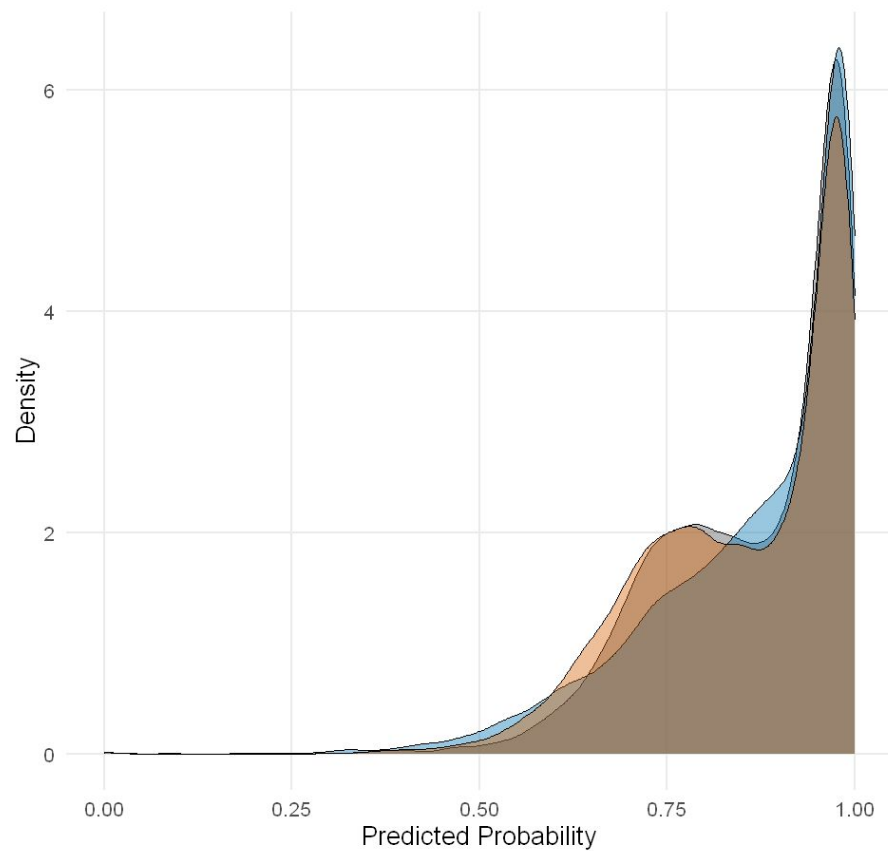
Model — M1_Baseline — M2_IPW — M3_Augmented

Bias Curve: Residual Analysis

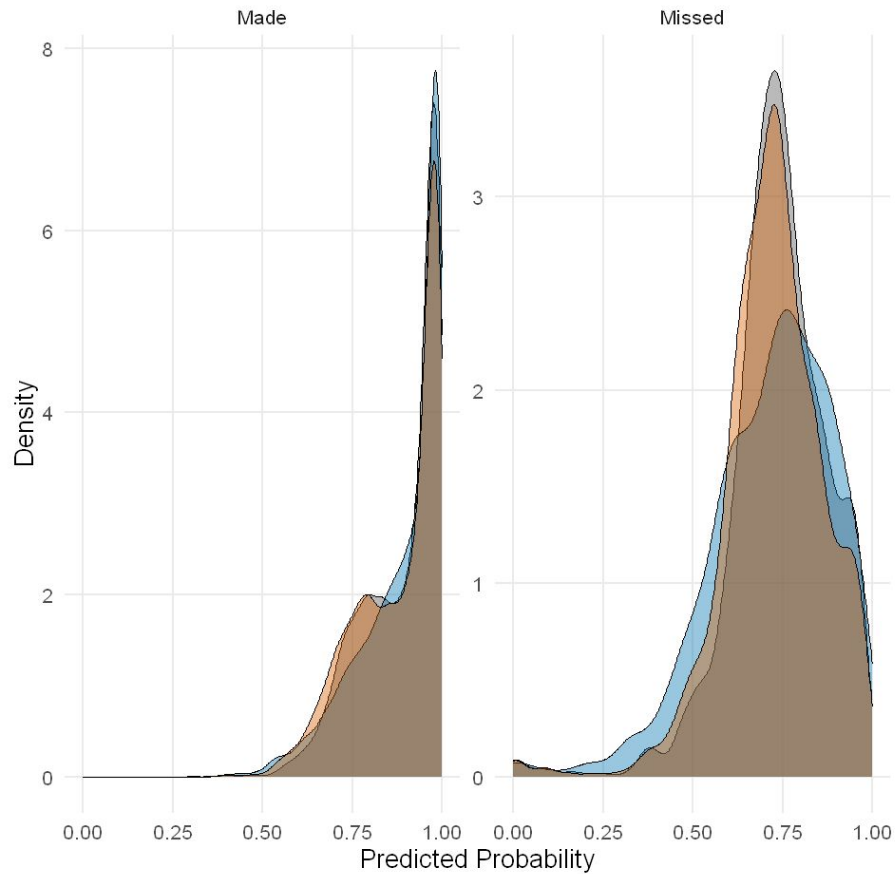


Model — M1_Baseline — M2_IPW — M3_Augmented

Global Probability Density



Separation (Make vs Miss)



Model M1_Baseline M2_IPW M3_Augmented

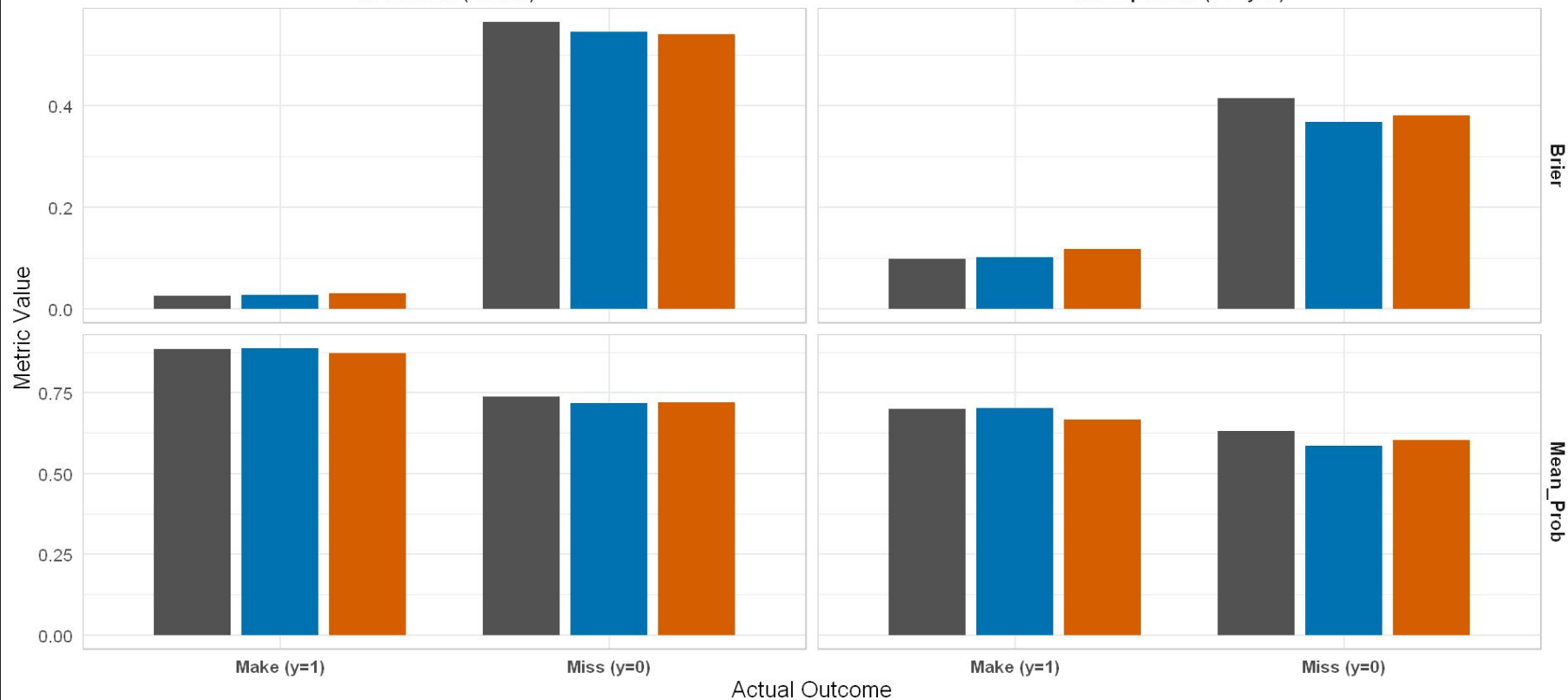
Model Diagnostics: Global vs. Deep Kicks

Comparing Brier Score and Predicted Probability on Makes vs. Misses

Model M1_Baseline M2_IPW M3_Augmented

1. All Kicks (Global)

2. Deep Kicks (50+ yds)



Metric Comparison (Out of Sample)

Subset	brier	auc	accuracy	precision	recall
M1 (Global)	0.098	0.772	0.876	0.879	0.995
M2 (Global)	0.100	0.760	0.872	0.882	0.986
M1 (50+ yds)	0.188	0.653	0.737	0.746	0.97
M2 (50+ yds)	0.193	0.641	0.717	0.752	0.912
M1 (Prob. 25-75%)	0.201	0.568	0.715	0.726	0.972
M2 (Prob. 25-75%)	0.206	0.571	0.697	0.731	0.917

Takeaways

Augmenting Field Goal Data Did NOT Improve Performance

IPW (Indirect - Simulating Decision Space)

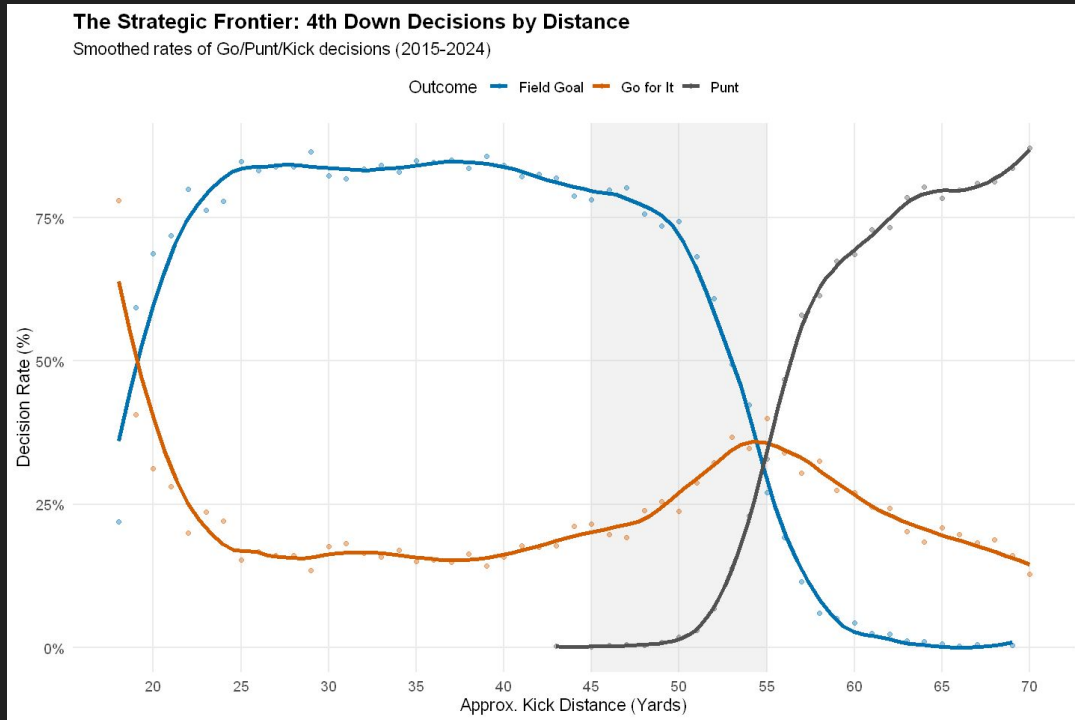
- **Performance:** Correcting for selection bias (IPW) provided **no significant global performance increase** (Brier/AUC). The "biased" reality (M1) is the most predictable one.
- **The 'Tail' Signal:** However, IPW *did* improve discrimination in the deep tail (50+ yards) by becoming more **pessimistic**. It learned that long kicks are harder (due to weights). Yet, overall calibration still worse than M1.

Direct Augmentation (PAT & Declined Attempts)

- **Performance:** Even at small weights (0.1) the impact was negative. Larger weights had greater impacts on probability distribution but as a result further worsened performance.
- "Hard Negative" lowers predicted probability for all kicks, does not help in differentiating makes vs. misses

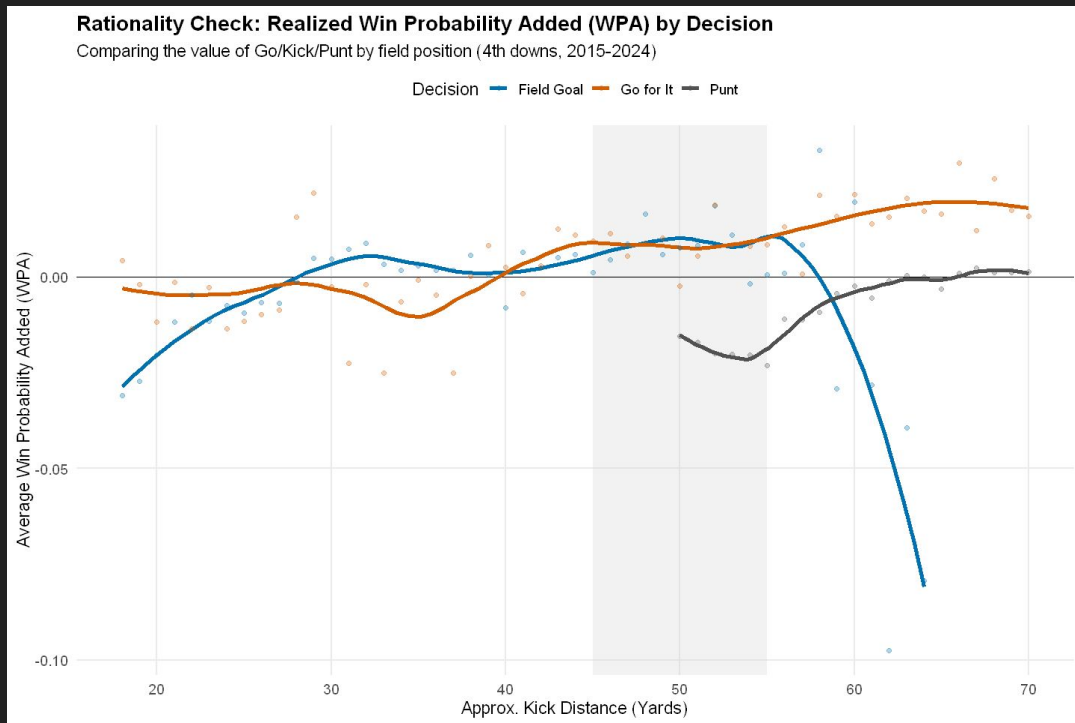
Takeaways

- **The Decision Zone:** The 45–55 yard range is "No Man's Land." This is where all play options converge
- Punting is inefficient (net ~20 yards), and Kicking is declining in value
 - $(x\text{Points} = 3 * \text{declining make } \%)$
- **Rationality Confirmed:** Coaches decline kicks in this zone **NOT** because they lack confidence in the kicker's leg strength, but because **maximizing Win Probability requires aggression.**
- **Modeling Implication:** Treating these non-attempts as "Misses" (Pseudo-Labeling) is structurally incorrect. It penalizes the kicker for the coach's correct strategic choice to chase a Touchdown.



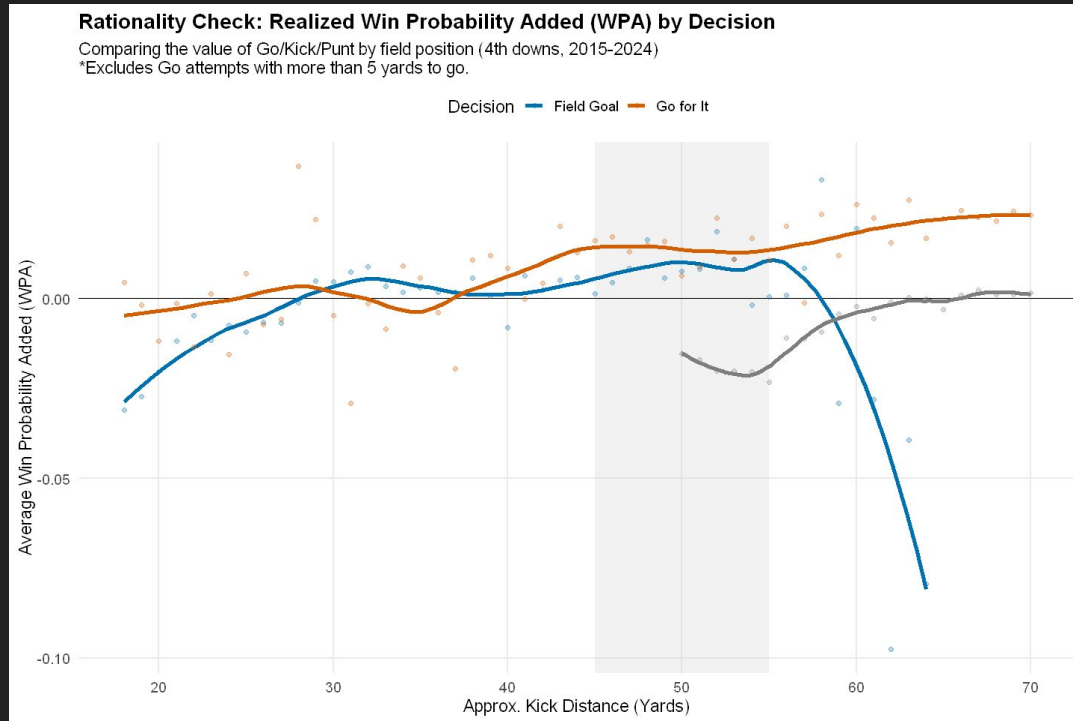
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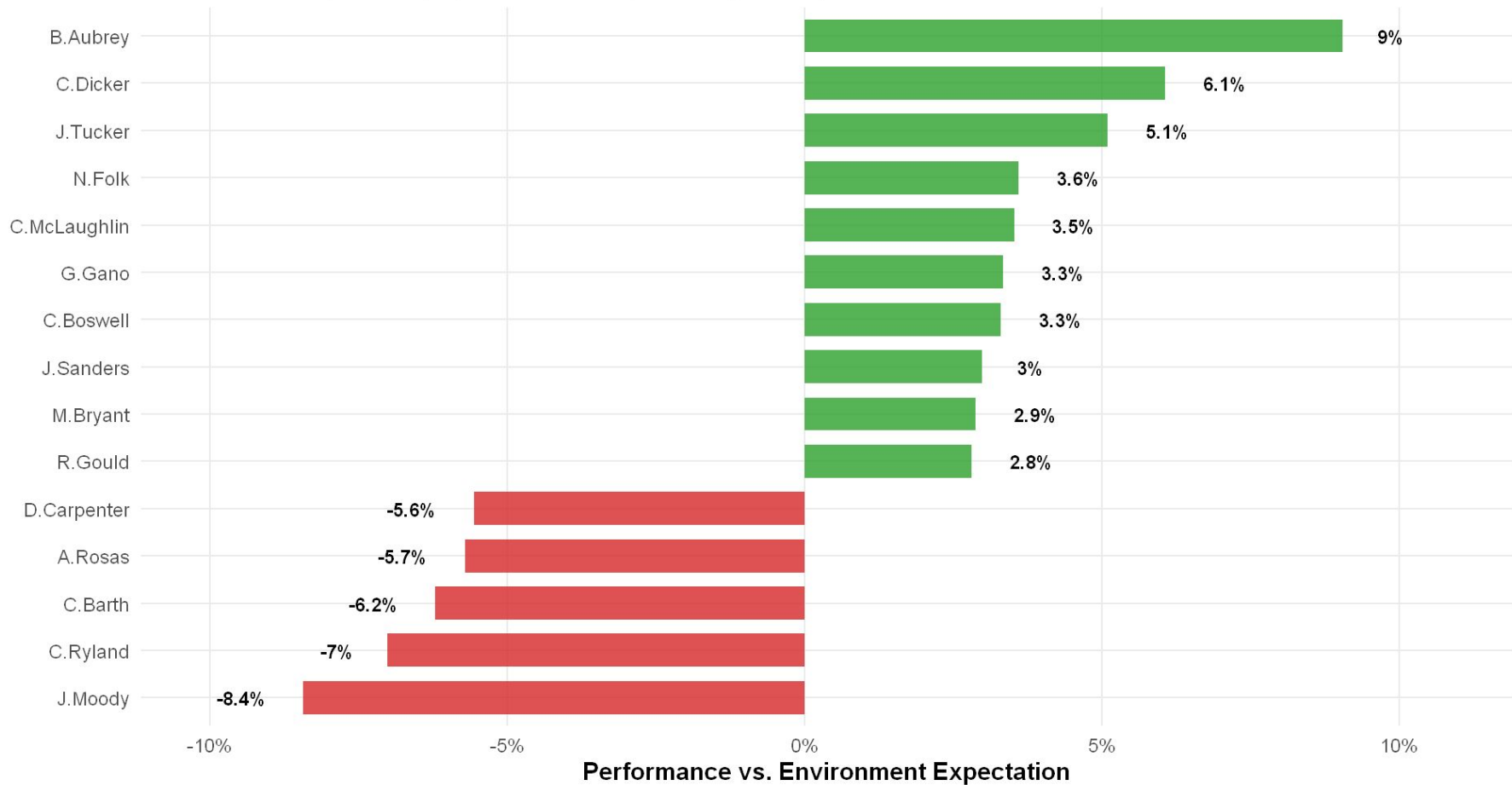
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Supplemental

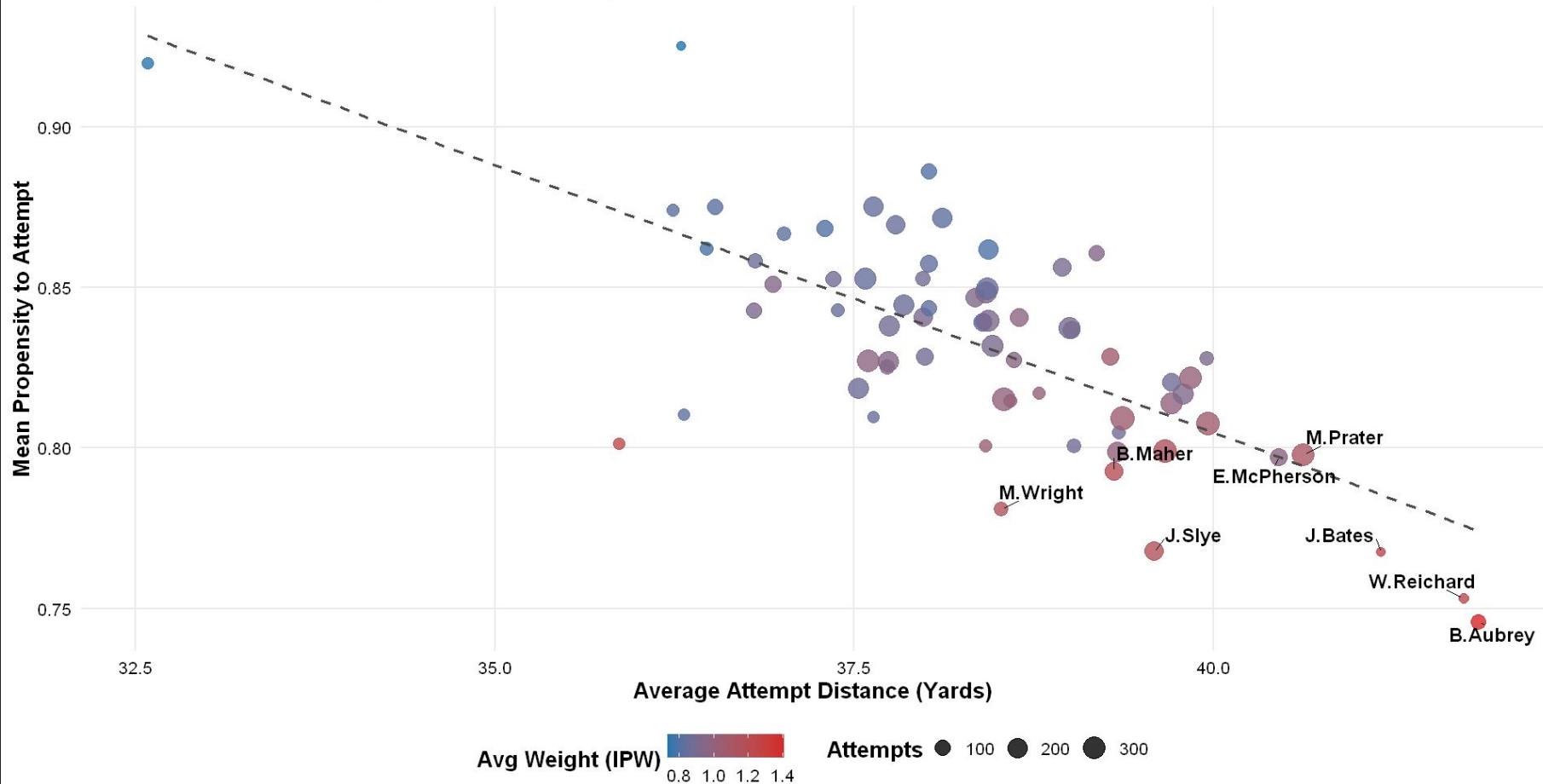
True Kicker Skill: FGOE Leaderboard (Kicker Agnostic Model)

Field Goal Over Expectation (Adjusted for Stadium, Weather, Season)



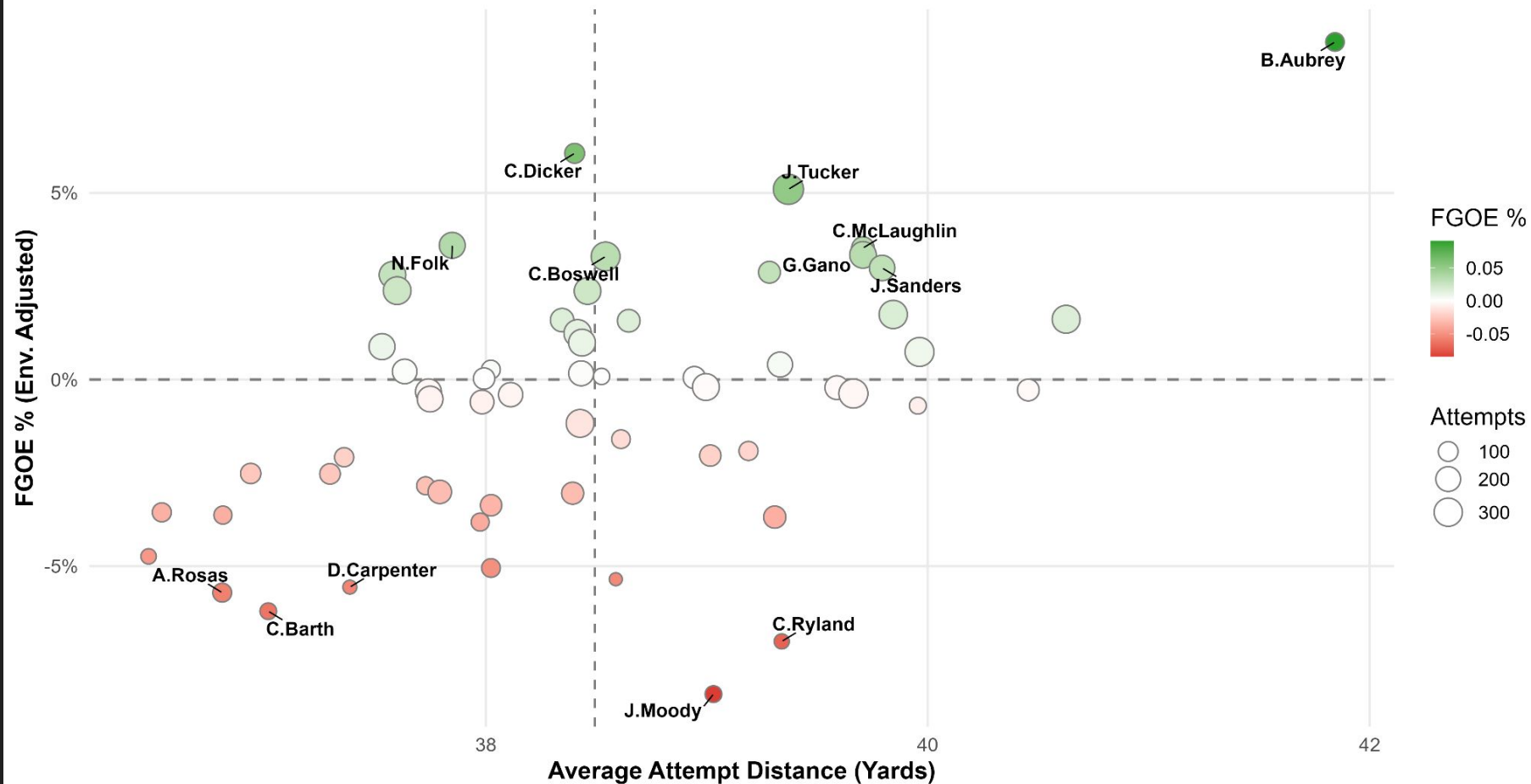
Kicker Workload Analysis

Who takes the non-standard kicks? (Lower Y = Harder Workload)



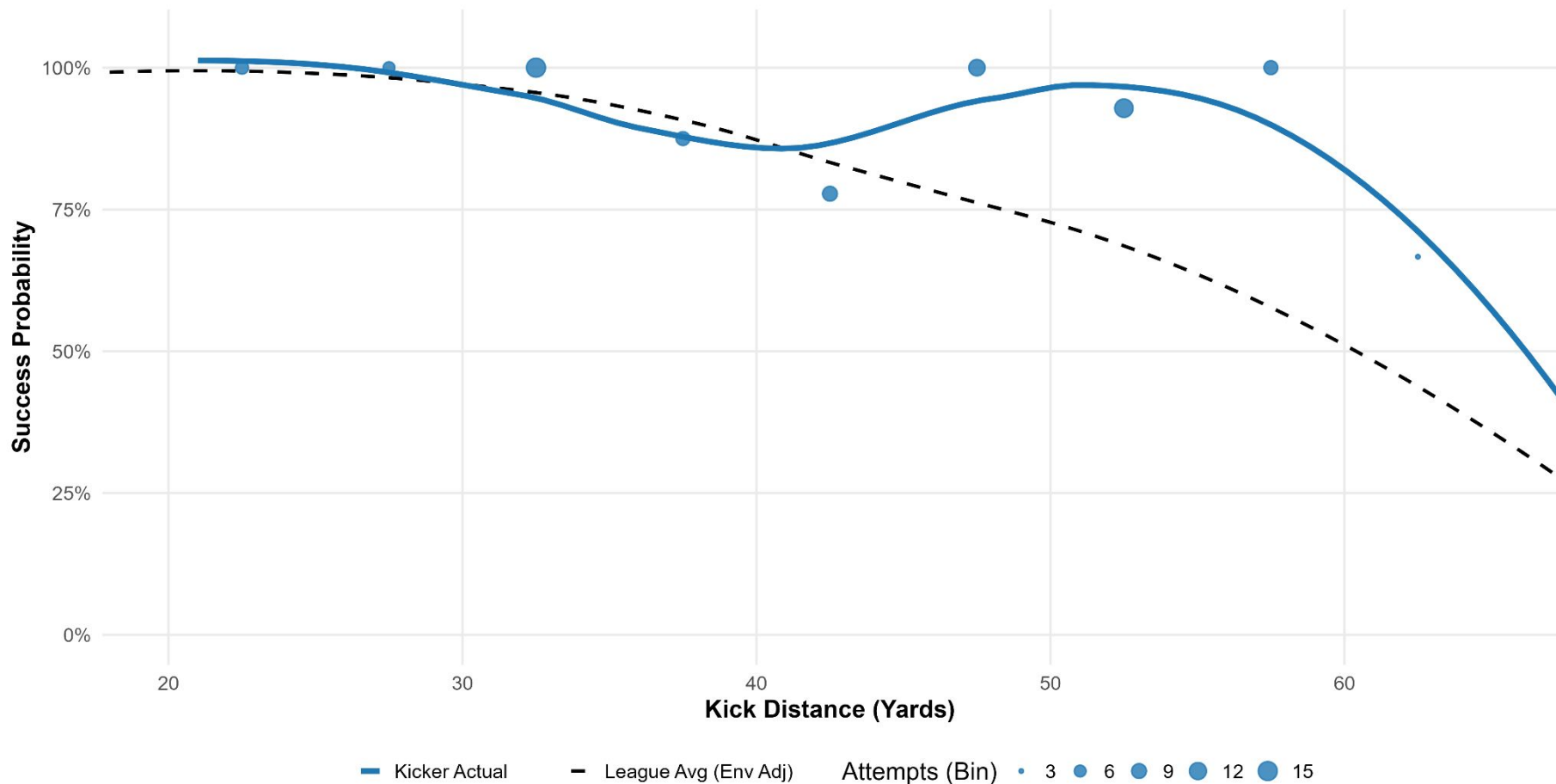
Skill vs. Workload Matrix

Does high skill correlate with difficult assignments?



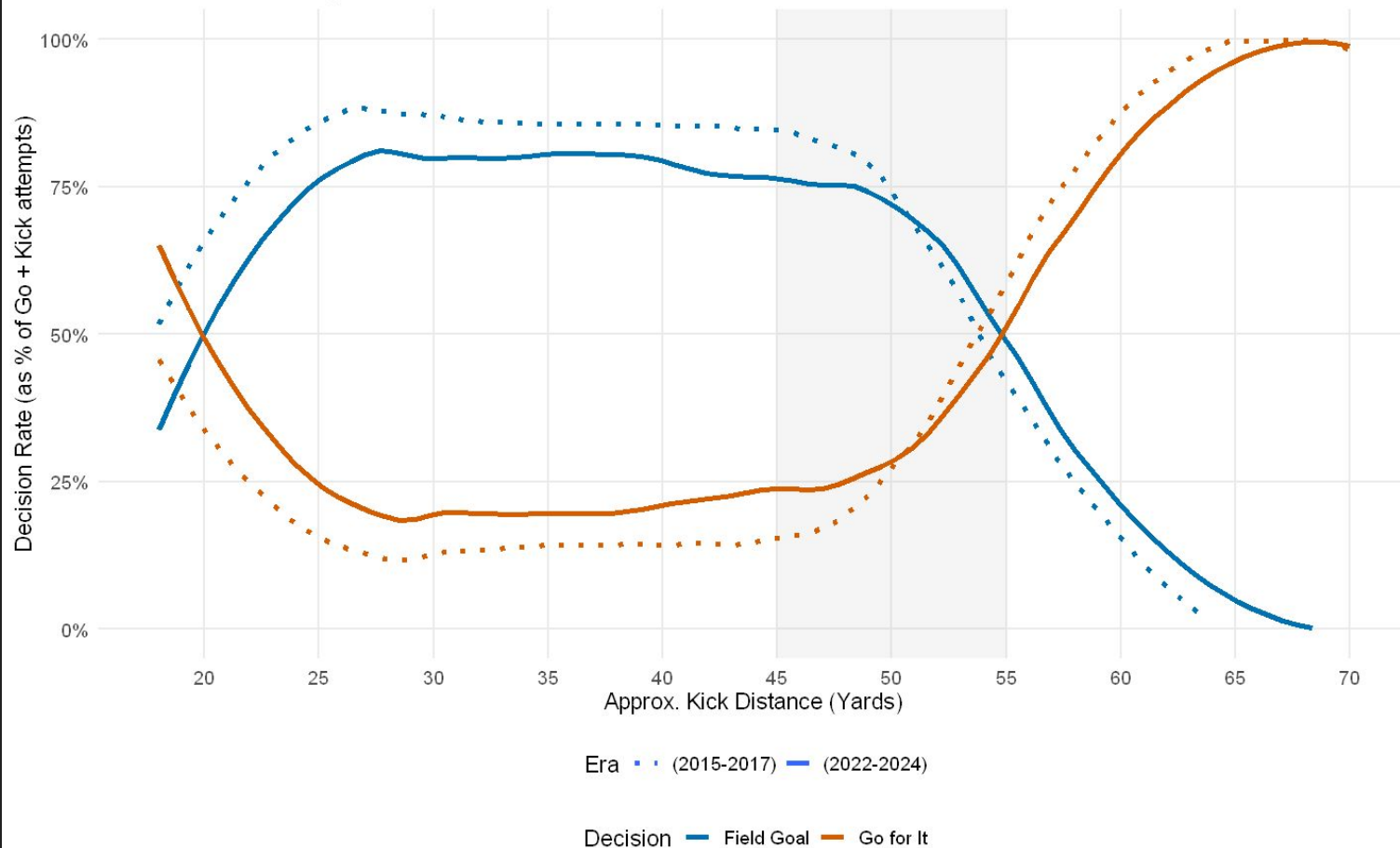
The Skill Gap: B.Aubrey

Actual Performance vs. Environment-Adjusted Expectation (FGOE: 9%)



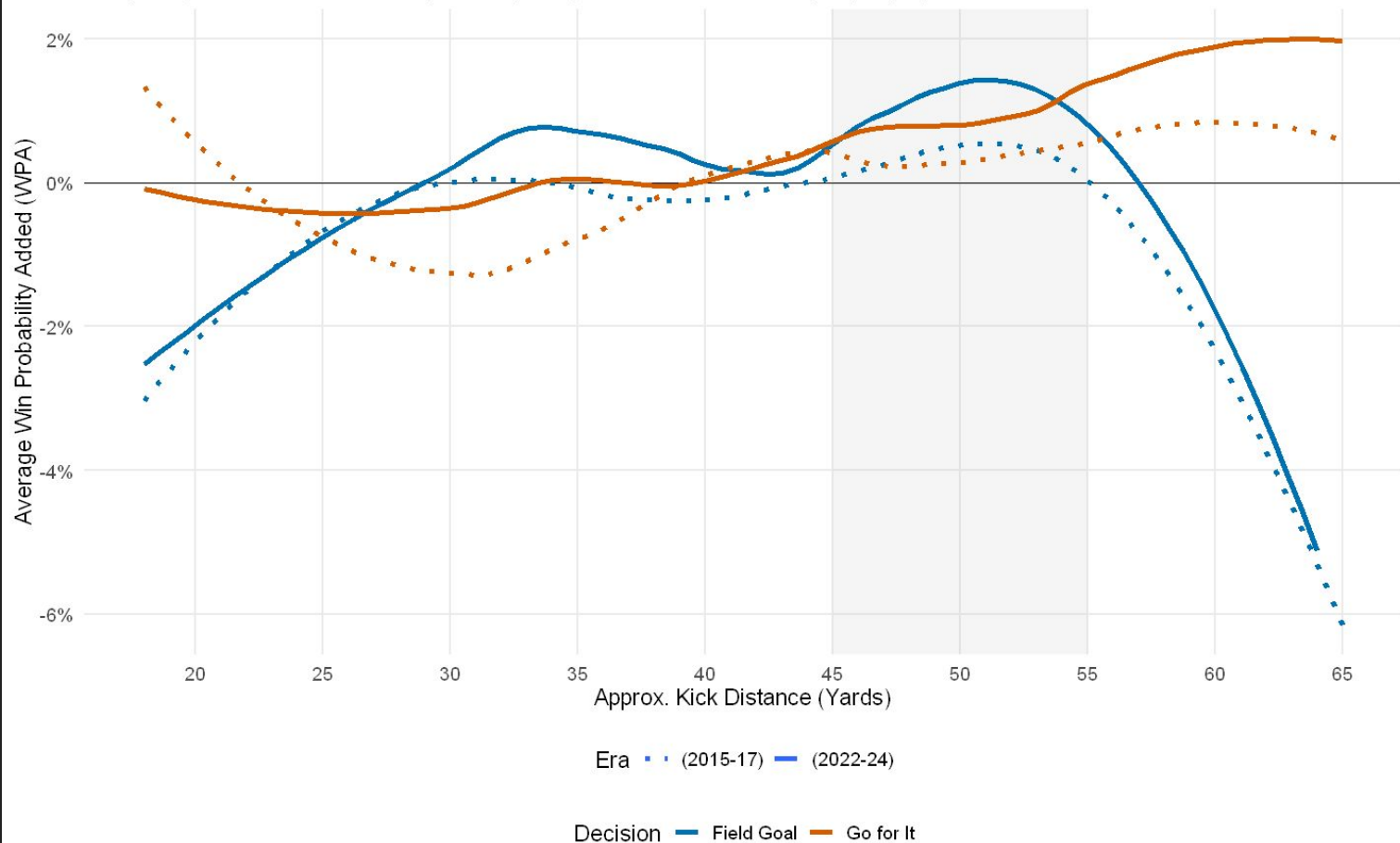
The Strategic Shift: Go vs. Kick (Conditional)

Decision rates excluding Punts.



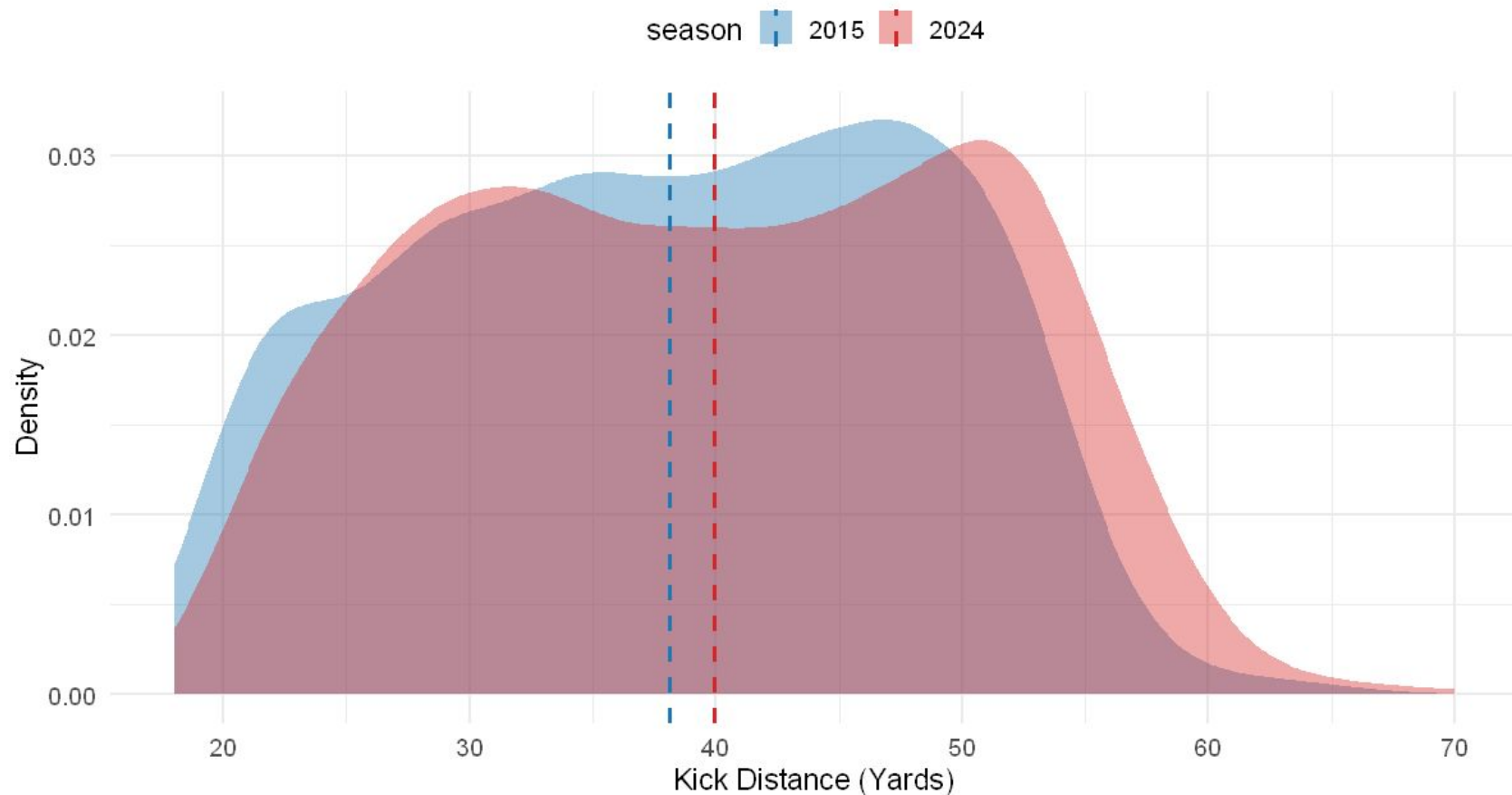
Rationality Check: WPA by Decision & Era

Comparing Realized Win Probability Added (WPA) for Kicks vs. 'Go' Attempts (<5 yds)



Shift in Kicking Strategy: 2015 vs. 2024

Distribution of Field Goal Attempt Distances. Dashed lines indicate mean distances.



Takeaways

- **The 'Unicorn' Effect:** The decreased IPW predicted probabilities were NOT largely driven by "desperation" (end-of-game heaves), as the propensity model already accounts for these. It was driven by "**Unicorn**" **attempts**—rare, long-distance kicks taken in standard game states (e.g., Q1/Q2) where coaches typically punt or go for it.

Season	Team	Dist	Score Diff	Qtr	Time Rem.	Weight	Made
2017	NO	56	+14	4	9:02	8.55	0
2017	BAL	62	-19	2	30:02	8.55	0
2017	ATL	59	+17	2	30:02	8.55	0
2017	DEN	53	-14	2	41:10	8.55	1
2017	NE	62	+14	2	30:05	8.55	1
2017	GB	57	+7	3	19:20	8.55	0
2017	LA	63	+7	2	30:02	8.55	0
2024	DAL	65	-7	1	51:38	8.17	1
2024	MIN	58	+24	4	4:48	8.17	1
2024	KC	65	-3	2	30:04	8.17	0

Inverse Probability Weighting Method

Objective: Correct selection bias via Inverse Probability Weighting (IPW) while preventing variance explosion from rare "Unicorn" kicks.

1. Probability Clipping (Safety Floor)

- **Action:** Clip predicted probabilities to $[0.02, 0.98]$ for stability.

2. Stabilized IPW Calculation

- **Action:** $w_{\text{raw}} = \text{Training Prevalence (FG)} / p_{\text{hat_clipped}}$
- **Why:** Uses the 3-year trailing average of FG attempts (the "Training Prevalence") as the numerator. This centers weights near 1.0, unlike standard IPW ($1/p$) which scales to sample size.

3. Season Normalization (Hájek Pass 1)

- **Action:** Scale weights so mean = 1 **within each season**.
- **Why:** Accounts for era effects. Ensures that a season with high attempt rates doesn't dominate the model simply because the weights drifted upwards.

4. Outlier Control (3-Sigma Cap & Re-Norm)

- **Action:** Cap weights at mean + 3 SD, then **re-normalize** to mean = 1.
- **Why:** The Cap stops extreme outliers from wrecking standard errors. The Re-Normalization restores the total weight mass, ensuring our Effective Sample Size (ESS) remains valid.

Inverse Probability Weighting Method

1. Probability Clipping (Safety Floor)

$$w_{i,\text{raw}} = \begin{cases} \frac{\bar{p}_{\text{target}}}{\hat{p}_{i,\text{clipped}}} & \text{if Attempt} = 1 \\ \frac{1 - \bar{p}_{\text{target}}}{1 - \hat{p}_{i,\text{clipped}}} & \text{if Attempt} = 0 \end{cases}$$

2. Stabilized IPW Calculation

$$w_{i,\text{hajek1}} = \frac{w_{i,\text{raw}}}{\frac{1}{N_s} \sum_{j \in S} w_{j,\text{raw}}}$$

3. Season Normalization (Hájek Pass 1)

$$\mu_s = \text{Mean}(w_{\text{hajek1}}), \quad \sigma_s = \text{SD}(w_{\text{hajek1}})$$

$$C_s = \mu_s + 3\sigma_s$$

$$w_{i,\text{capped}} = \min(w_{i,\text{hajek1}}, C_s)$$

4. Outlier Control (3-Sigma Cap & Re-Norm)

$$w_{i,\text{final}} = \frac{w_{i,\text{capped}}}{\frac{1}{N_s} \sum_{j \in S} w_{j,\text{capped}}}$$

